VARIANCE AND EXPECTED VALUE

The same moves, done to E and to Var

Expectation passes through sums for any variables; variance only for independent ones.

✓ ALWAYS ◐ IF INDEPENDENT

MOVE	Expected value	Variance
A constant c	✓ ALWAYS $E[c] = c$	✓ ALWAYS $\mathrm{Var}(c) = 0$
Shift $X + b$	✓ ALWAYS $E[X + b] = E[X] + b$	✓ ALWAYS $\mathrm{Var}(X + b) = \mathrm{Var}(X)$
Scaling aX	✓ ALWAYS $E[aX] = a\,E[X]$	✓ ALWAYS $\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$
Sum $X + Y$	✓ ALWAYS $E[X] + E[Y]$, even if dependent	◐ IF INDEPENDENT $\mathrm{Var}(X) + \mathrm{Var}(Y)$
Product XY	◐ IF INDEPENDENT $E[X]\,E[Y]$	NOT ON THIS PAGE

Shifting moves the mean but not the spread; scaling by a multiplies the spread by a^2.